

IFT MENTORING MANUAL FOR EDUCATORS

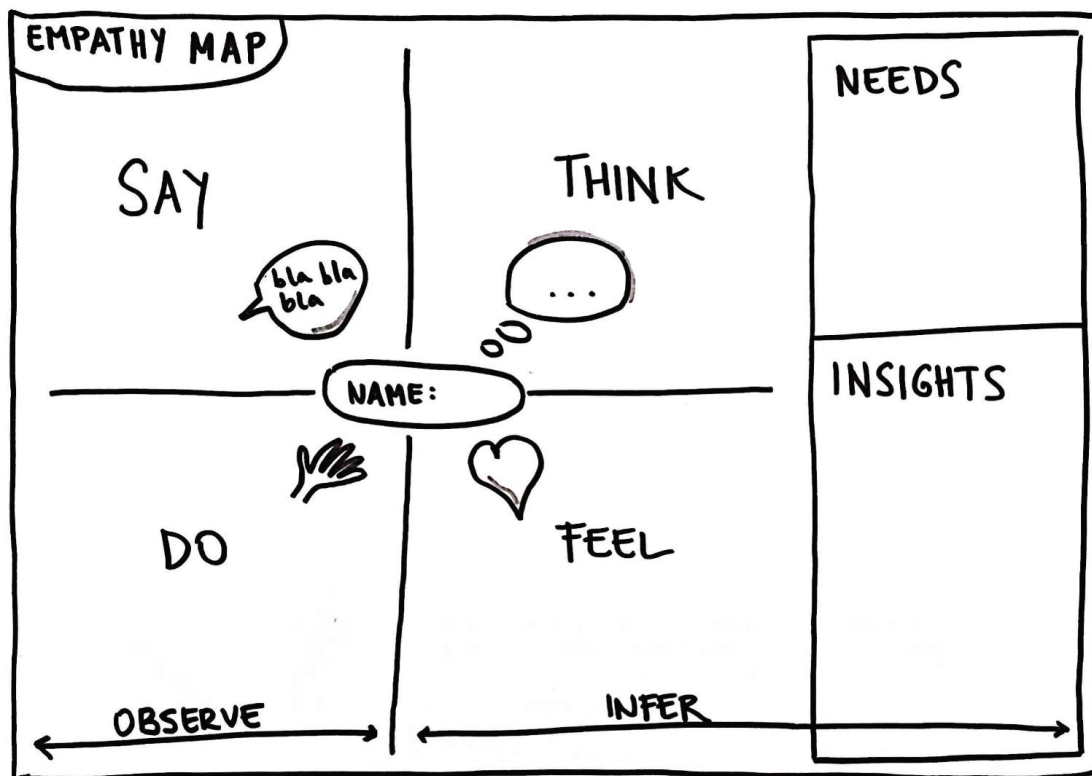
This Manual Serves As A Supplement To The Training Videos Available On The **IFT** Platform.



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01. Empathy Map



It is a structured tool to understand a user's mindset by capturing what they **Think, Say, Do,** and **Feel** about a product, process, or situation. The Empathy Map framework helps students step outside their own assumptions and see a problem or product from the real user's point of view, before jumping to solutions.

Why Use The Empathy Map Framework?

- **Builds User-Centered Thinking:**
Shifts focus from "what I assume" to "what the user actually experiences."
- **Uncovers Hidden Needs:**
Surfaces frustrations and desires users may not directly state.
- **Structured Empathy:**
Breaks down a person's experience into 4 clear, observable dimensions.
- **Strong Foundation for Ideation:**
Once real user pain points are visible, students can design solutions that actually matter.

How To Use The Empathy Map Framework?

1. Choose a User and a Situation:

Pick a real person (or persona) using a specific product or going through a specific process.
Example: A person using a coffee cup.

2. Ask the 4 Empathy Map Questions:

- **Think:** What questions, doubts, or thoughts run through the user's mind (often unspoken)?
Example: "How to keep coffee warm?" "Don't know how much is left?" "Glass cups often break."
- **Say:** What does the user actually say out loud about the product or situation? Example:
"How can I know if it's hot or cold?" "Can coffee drinking be fun?" "Easy to know what's/how much is there?"
- **Do:** What actions does the user actually take (including workarounds)? Example: "Keep checking the quantity," "Hold it with cloth," "Keep touching to see the temperature," "Keep warming in the microwave repeatedly."
- **Feel:** What emotions does the user experience during this situation? Example: "Painful to get up for heating," "Scared while travelling with coffee," "Boring to see the same white cup," "Anxious while touching the cup."

3. Document Insights:

Place all answers into the four quadrants around the persona, so the full picture of the user's experience is visible at a glance.

4. Spot the Gaps and Pain Points:

Look for places where Think, Say, Do, and Feel don't match - these gaps often reveal the real problem. Example: The user Says "how can I know hot/cold?" but Does "keep touching to see the temperature" - showing there's no easy way to check without physical risk.

5. Brainstorm Solutions:

Use the pain points found across all four quadrants to design a product or fix that addresses the real, felt experience - not just the obvious one.

How Can Educators Use The Empathy Map Framework?

• Explain Each Quadrant with Examples:

Use a simple, relatable object (like a coffee cup) before moving to the students' own topic.

• Encourage Real Observation:

Ask students to actually watch or interview someone using the product, not just guess.

• Separate Say from Think:

Help students notice that what people say out loud is often different from what they're actually thinking or worried about.

• Guide Emotion Vocabulary:

Some students struggle to name feelings - give them a short list of emotion words (anxious, bored, scared, frustrated, relieved) to choose from.

Promote Group Work:

Let groups interview different users of the same product/process so multiple empathy maps can be compared.

Troubleshooting Points For Educators

1. Students confuse "Think" and "Say"

Students often put the same sentence under both, not realizing Think is private/unspoken while Say is what's spoken aloud. Example: A student may write "How can I know if it's hot?" under both boxes, when it may only be a spoken complaint (Say), while the private worry (Think) is "I don't know how much coffee is left." Fix: Explain: **"Think"** is what's in their head but they'd never say out loud. **'Say'** is what they'd actually tell a friend." Ask, "Would this person really say this sentence to someone?"

2. Students confuse "Do" with "Feel"

Students describe an action but label it as a feeling, or vice versa. Example: Writing "scared" under Do, instead of the actual physical action like "holds cup with cloth" (Do) which is caused by the feeling "scared to get burnt" (Feel). Fix: Ask, "Is this something you can see them physically doing, or something happening inside them emotionally?" Do = visible action, Feel = emotion.

3. Students only fill in one or two quadrants.

Students find "Feel" or "Think" harder to imagine than "Do" or "Say," so they leave them empty or rushed. Example: A student may list 4-5 things under Do, but only 1 vague word under Feel. Fix: Set a minimum requirement (e.g., at least 3 points per quadrant) and prompt directly: "How would this make YOU feel if it happened to you?"

4. Students describe the product, not the person's experience.

Instead of empathizing with the user, students describe features of the object itself. Example: Writing "the cup is made of glass" under Think, instead of "worried the glass cup might break" (the person's actual thought/fear). Fix: Remind students every point must relate to a person's mind, words, actions or emotions — not a fact about the object.

5. Students write generic, one-word answers.

Short answers like "happy" or "confused" don't give enough insight to design around. Example: "Confused" under Feel, instead of "Anxious while touching cup in case it's too hot." Fix: Ask a follow-up for every entry: "Confused about what, exactly? When does this happen?" Push for a full sentence tied to a specific moment.

6. Students don't connect the four quadrants into one user story.

Students treat each quadrant as isolated, missing how they link together to reveal a real pain point. Example: Not noticing that "keeps touching to check temperature" (Do) connects directly to "scared while travelling with coffee" (Feel) and "how can I know hot/cold?" (Think/Say) — all pointing to the same core problem: no way to safely gauge temperature. Fix: After filling all four boxes, ask students to circle 2-3 points across different quadrants that seem connected, and explain the link out loud.

Tips For Using The Empathy Map Framework

- **Use a Real Persona:**
Give the user a face, name, and short backstory — it makes empathy easier than analyzing an abstract "user."
- **Color-Code the Quadrants**
Keep Think, Say, Do, and Feel visually distinct (as in the reference slide) so patterns are easy to spot at a glance.
- **Interview Before Mapping:**
Where possible, have students briefly interview or observe a real person before filling the map.
- **Look for Mismatches:**
The most valuable insights often come from where Say and Do don't match — that gap usually points to the real problem.

Benefits For Students

- **Builds Genuine Empathy:**
Moves students beyond assumptions to understanding how a real user thinks and feels.
- **Improves Observation Skills:**
Trains students to notice subtle cues in behavior and language, not just obvious statements.
- **Strengthens Problem Framing:**
Provides richer, emotion-backed insights that lead to more meaningful solutions later in the design process.
- **Encourages Perspective-Taking:**
Helps students practice seeing a situation from someone else's point of view - a core life skill beyond just entrepreneurship.

02. Problem Identification

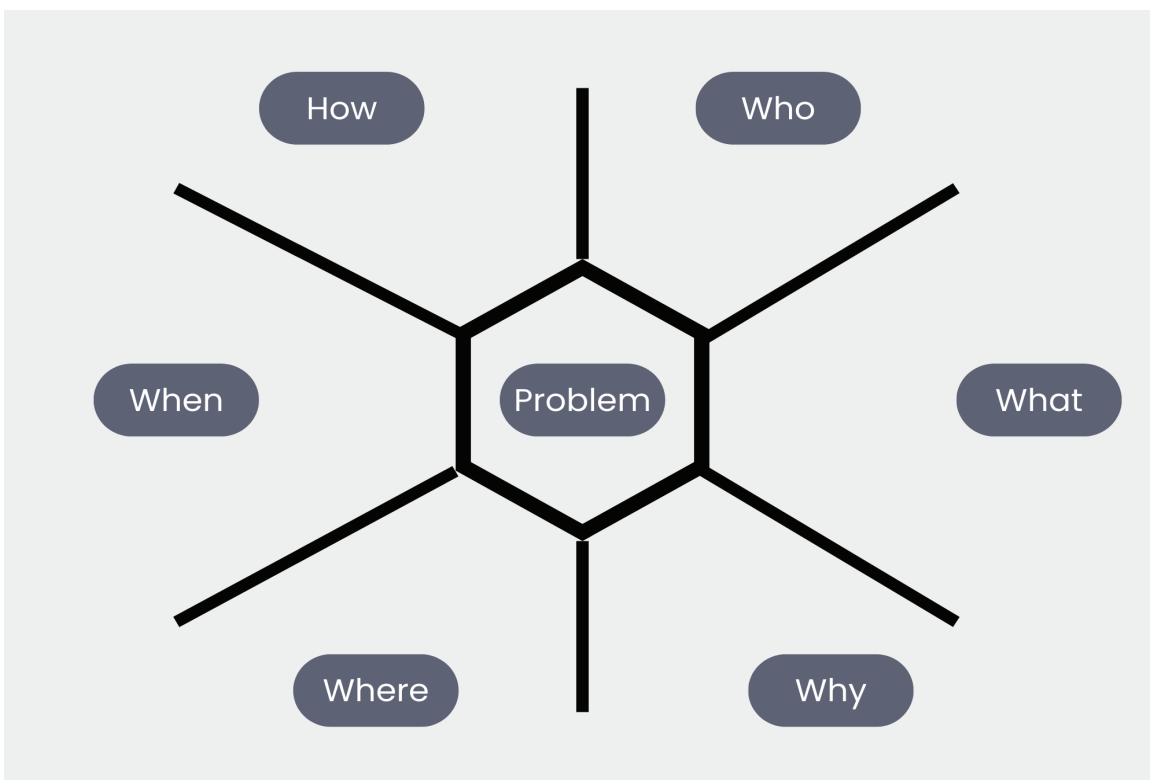
5W1H Framework

It is a structured tool to identify and analyze the problem, its impact, and the affected audience. The 5W1H framework is a systematic approach to analyzing a problem or situation by asking six fundamental questions: **Who, What, When, Where, Why, and How.** This method provides a 360-degree view of the issue and helps identify all relevant aspects before working on solutions.

Why Use The 5W1H Framework?

- **Thorough Analysis:**
Encourages students to explore the problem from multiple angles.
- **Structured Thinking:**
Breaks down the problem into manageable parts.
- **User-Centered:**
Helps identify key stakeholders and their needs.
- **Action-Oriented:**
Encourages practical steps for problem-solving.

Visual Representation Of 5W1H Framework:



How To Use The 5W1H Framework?

1. Start with the Problem:

Clearly state the problem in the center of the framework (as shown in your diagram).

2. Ask the 5W1H Questions:

Use the following questions as guiding prompts to analyze the problem: Eg: Rural children miss school due to recurring sickness

Who:

- Who is affected by this problem?
- Who is responsible for causing or solving it?
- Example: Families in the village are affected by dirty water.

What:

- What is the problem?
- What are its effects or symptoms?
- Example: Dirty water causes health issues like stomach infections.

When:

- When does the problem occur?
- Is it linked to specific times, seasons, or events?
- Example: The problem worsens during the summer when water availability decreases.

Where:

- Where does the problem occur?
- Is it specific to certain locations?
- Example: The issue is prevalent in rural areas with limited water resources.

Why:

- Why is this problem happening?
- What are the root causes?
- Example: There's no access to affordable water filtration methods.

How:

- How does this problem affect people or processes?
- How can it be addressed?
- Example: Families suffer from poor health, and children miss school due to sickness.

3. Document Insights:

Use the answers to create a detailed picture of the problem and its context.

4. Brainstorm Solutions:

With a thorough understanding, students can now think of specific solutions to tackle the identified problem.

How Can Educators Use The 5W1H Framework?

• Explain the Questions:

Break down each question with examples relatable to students.

- **Encourage Observation:**

Ask students to observe their surroundings to answer these questions.

- **Guide Brainstorming:**

Help them connect the answers to form a clear understanding of the problem.

- **Promote Collaboration:**

Allow students to work in groups to answer the questions, encouraging diverse perspectives.

Troubleshooting Points For Educators

1. Students mix up "What" and "Why"

Students often describe the problem again when asked "Why," instead of finding the root cause. Example: For "Rural children miss school due to sickness," a student may answer Why as "Children get sick" (this is What, already covered) instead of "Because drinking water is not filtered properly" (this is the real cause). Fix: Tell students — What describes the problem. Why explains the reason behind it. Ask them, "If nothing is done, why will this keep happening?"

2. Students give very short, one-word answers

like "Village" for Where, or "Students" for Who, don't give enough detail to work with. Example: Instead of just "Village," a fuller answer is "Government schools in villages that don't have a piped water supply." Fix: Ask a follow-up question for each answer — "Can you tell me more? Which village? Why that one?" Give a target of at least one full sentence per answer.

3. Students confuse "How" as a solution step, not an impact/method question

Since How comes last, students sometimes jump straight to "We will build a filter," skipping the analysis of how the problem currently affects people. Example: How should first capture "Sick children miss 3-4 school days a month and fall behind in class," before any solution is proposed. Fix: Explain that How has two parts — first, how the problem affects people now; solutions come only in the next brainstorming step.

4. Students only think from one point of view.

Groups often describe the problem only as they personally see it, missing other people involved. Example: For the water problem, students may talk only about children, forgetting parents who lose wages caring for sick kids, or Educators who must manage attendance gaps. Fix: Ask "Who else is affected, even a little?" Prompt them to think of at least 3 different people or groups for the Who question.

5. Students treat all 6 questions as separate, unconnected answers.

They fill in each box individually without seeing how the answers connect into one clear problem picture. Example: Their Who, What, and Why answers may not logically link — Who says "farmers," but Why talks about "school-going children," creating a disconnect. Fix: Once all 6 are filled in, ask students to read their answers aloud as one connected paragraph or story. If it doesn't make sense together, they need to revisit the mismatched answers.

6. Students confuse symptoms with root causes across multiple questions.

The same root-cause confusion from Why often repeats in What (symptoms vs. problem) and How (effects vs. deeper impact). Example: "Children are absent" is a symptom; "There is no clean water source nearby" is the actual problem. Fix: Teach a simple test - ask "Is this something we can observe directly (a symptom), or something that needs digging to find (a cause)?" Use the analogy of a fever (symptom) vs. an infection (cause)

Tips For Using The 5W1H Framework

- **Be Specific:**
Encourage detailed answers to each question.
- **Use Visual Aids:**
Students can use charts, diagrams, or sticky notes to organize their answers.
- **Encourage Curiosity:**
Motivate students to ask follow-up questions for deeper insights.
- **Focus on Actionable Answers:**
Ensure that answers lead to practical and realistic solutions.

Benefits For Students

- **Improves Critical Thinking:**
Encourages analysis of problems from multiple angles.
 - **Enhances Problem-Solving Skills:**
Provides a structured method to tackle complex issues.
 - **Promotes Collaboration:**
Allows students to brainstorm and discuss answers collectively.
 - **Boosts Confidence:**
Helps students gain clarity and a deeper understanding of the problem.
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04. Activity Analysis

It is a structured tool to break down a process into its individual steps and analyze what is happening, why it is happening, how it is happening, when, where, and how much effort/cost is involved at each step. The Activity Analysis framework helps students look closely at a familiar process and uncover hidden problems, inefficiencies, or gaps that they may never have noticed before.

Why Use The Activity Analysis Framework?

- **Deep Observation:**
Trains students to look beyond the surface of a process and notice details they usually overlook.
- **Step-by-Step Thinking:**
Breaks a large, complex process into smaller, manageable stages.
- **Root-Cause Discovery:**
Helps students see not just what is happening, but why it is happening at each stage.
- **Foundation for Ideation:**
Once problems are identified stage-by-stage, students have solid ground to brainstorm real solutions.

How To Use The Activity Analysis Framework?

1. **Choose a Process:**
Pick a real, everyday process that students can relate to or observe. Example: Garbage collection process in the city.
2. **Break the Process into Steps:**
List out every stage the process passes through, in order. Example: In the house → Garbage collector at doorstep → Drum to common bin → Common bin to truck → Truck to dumping ground → Scavenging → Separation for recycling → Various recyclers.
3. **Analyze Each Step Using These Questions:**
 - **What is happening?**
Describe what actually takes place at this step. Example: At the "common bin to truck" stage — spillage, mixing of wet and dry waste, constant smell, animals rummage through it, pests and insects spread.
 - **Why is it happening?**
Dig into the root causes behind what's happening. Example: Mixing of waste happens because of laziness, not every room has a dustbin, the maid mixes it up, or people simply aren't aware they need to separate waste.

- **How is it happening?**

Identify whether the step is manual, semi-automatic, or automatic. Example: Waste collection at the doorstep is manual; truck loading is semi-automatic.

- **When does it happen?**

Note the timing or frequency of the step. Example: House to doorstep collection mostly happens in the morning but continues through the day.

- **Where does it happen?**

Pin down the exact location of the step. Example: Within the house, in the corridor, at the building compound, or at the dumping yard.

- **How much effort/cost is involved?**

Estimate the physical effort, money, or hidden costs (including health/environmental costs). Example: At the dumping ground, tremendous environmental and health costs are involved — diseases spread that cannot even be measured in money.

4. Document Insights:

Record the answers for every step in a simple table so the entire process is visible at a glance.

5. Identify the Problem Areas:

Look across all steps to spot where the biggest issues, inefficiencies, or risks are concentrated.

6. Brainstorm Solutions:

Once problem areas are clear, move to solution-building for the most critical stages.

How Can Educators Use The Activity Analysis Framework?_____

- **Model One Step First:**

Walk through one step of the process together as a class before asking students to do the rest independently.

- **Encourage Real Observation:**

Push students to think about what they've actually seen, not just what they assume happens.

- **Guide Depth in "Why":**

Don't let students stop at the first reason — ask "why" again to dig deeper.

- **Promote Group Work:**

Have small groups each take a few steps of the process so the full analysis comes together faster.

- **Connect the Dots:**

Help students see how a problem at one step (e.g., overfilled bins) causes a bigger problem downstream (e.g., trucks rocking and spilling waste).

Troubleshooting Points For Educators

1. Students describe the process instead of analyzing each step.

Students often just narrate what happens ("the truck comes and picks it up") without answering what, why, how, when, where separately. Example: For "truck to dumping ground," a student may only write "the truck goes to the dumping ground" instead of noting the spillage, smell, and cost involved. Fix: Give students a simple table with the 6 question-columns and insist every box gets its own answer, even if short.

2. Students stop at the first "why" without digging deeper.

Students give a surface-level reason and stop, missing the real root cause. Example: A student may write "mixing happens because people are lazy," without going further to note that "every room doesn't have a dustbin" is also a real cause. Fix: Ask "Is that the only reason, or is there something behind that reason too?" Push for at least 2-3 causes per step.

3. Students confuse "How" (method) with "Why" (reason).

Since both questions can sound similar, students often repeat their "Why" answer under "How." Example: Under How, a student may write "because people don't separate waste" (this is a Why) instead of "manually" or "semi-automatically" (this is a How). Fix: Clarify that How asks "By what method does this happen — by hand, by machine, or both?" not "What caused it?"

4. Students skip steps they find boring or hard to observe.

Steps that are less visible to daily life (like "scavenging" or "separation for recycling") often get left blank or rushed. Example: Students living in apartments may not know that scavengers manually pick through waste at the dumping ground and skip this step entirely. Fix: Encourage students to ask family, watch videos, or research briefly for steps they haven't personally observed, rather than skip them.

5. Students underestimate "effort/cost,"

especially hidden costs. Students think of cost only as money and miss effort, time, health, or environmental costs. Example: A student may write "no cost" for the dumping ground stage, missing that diseases spreading and environmental damage are real, serious costs. Fix: Explain that cost isn't just rupees — ask "Does this step hurt anyone's health, waste anyone's time, or damage the environment?"

6. Students treat every step as equally important.

Without guidance, students may spend equal time on a minor step and a major problem step, losing focus on where the real issue lies. Example: Spending as much time analyzing "in the house" (low problem) as "common bin to truck" (high problem, lots of spillage and disease risk). Fix: Once all steps are filled in, ask students to mark which 2-3 steps have the most serious issues, and focus deeper discussion there.

Tips For Using The Activity Analysis Framework

- **Use a Table Format:**

A step-by-step grid (steps as rows, questions as columns) keeps the analysis organized and easy to compare.

- **Encourage Real-World Research:**

Students can ask family members, watch short videos, or observe directly to fill in gaps.

- **Look for Patterns:**

Encourage students to notice if the same root cause (e.g., "lack of awareness") repeats across multiple steps.

- **Keep it Visual:**

Diagrams or flowcharts of the steps help students see the process as a journey, not just a list.

Benefits For Students

- **Builds Observational Skills:**

Students learn to notice details in everyday processes they usually take for granted.

- **Strengthens Root-Cause Thinking:**

Moves students beyond surface symptoms to real underlying causes.

- **Prepares for Problem-Solving:**

Gives students a clear, evidence-based starting point for brainstorming solutions.

- **Encourages Empathy:**

Understanding the effort and hidden costs at each step (like a scavenger's health risk) builds empathy for people involved in the process.

05. Defining a Problem & Problem Identification Canvas

It is a structured process to convert raw research (Activity Analysis + Empathy Map insights) into one clear, well-written problem statement, and then examine that problem from every angle using a 6-box canvas - before any solution is attempted. This step ensures students solve the real problem, not just the first symptom they notice.

Why Use This Framework?

- **Converts Research into Clarity:**
Turns scattered observations (Who, What, Why, insights) into one focused, actionable problem statement.
- **Prevents Solving the Wrong Problem:**
Forces students to identify root cause, not just symptoms, before ideating.
- **360-Degree View:**
The canvas covers context, customers, emotional and quantifiable impact, and existing alternatives - so no angle is missed.
- **Sets Up Strong Ideation:**
A well-defined problem, backed by a canvas, gives students a solid, evidence-based launchpad for brainstorming solutions.

How To Use This Framework?

Part 1: Write the Problem Statement

1. Recall Who, What, Why:

Go back to the earlier 5W1H / Activity Analysis work and pull out the Who (user), What (problem), and Why (root cause).

2. Convert Into a Problem Statement Using 3 Parts:

- **<User>** — Who exactly is affected?
- **<Need>** — What do they need (not the solution, just the need)?
- **<Insight>** — Why does this need exist (the root cause/insight behind it)?

3. Example, using the waste management problem:

- **User:** Families living in areas without a formal waste segregation system.
- **Need:** A simple way to separate wet and dry waste at home, every single day.
- **Insight:** Because there's no awareness of why segregation matters, and not every room has a dustbin, waste gets mixed from the very first step — making every step after it (collection, transport, disposal) messier, smellier, and more harmful.

4. Write it as One Sentence:

Combine the three parts into a single, clear problem statement. Example: "Families in areas without a segregation system need a simple way to separate wet and dry waste at home, because low awareness and missing dustbins cause mixing right from the first step."

Part 2: Build the Problem Identification Canvas

5. Fill the 6 Boxes of the Canvas:

- **Context - When does the problem occur?**

Example: Every day, right when waste is generated in the house, and worsens during monsoons.

- **Problem - What is the root cause of the problem?**

Example: No awareness of the need to segregate, and no dustbin in every room.

- **Alternatives — What do people do now to fix the problem?**

Example: Some households buy an extra bin; some just leave it to the maid/domestic help to sort.

- **Customers — Who has the problem most often?**

Example: Families without a household help dedicated to sorting, or where segregation habits were never built in.

- **Emotional Impact & Quantifiable Impact — How does it feel, and what's the measurable cost?**

Example (Emotional): Frustration, embarrassment when garbage smells in the corridor.

Example (Quantifiable): X kg of recyclable waste lost to landfill per household per month.

- **Alternative Shortcomings — What's wrong with the current fixes?**

Example: An extra bin doesn't work if no one explains why segregation matters; relying on domestic help fails on their day off.

5. Cross-Check the Canvas Against the Problem Statement:

Make sure everything filled into the 6 boxes supports and matches the User-Need-Insight statement written in Part 1.

6. Move to Ideation:

With a validated, well-rounded problem definition, students are now ready to brainstorm solutions.

How Can Educators Use This Framework?

- **Model the User-Need-Insight Format First**

Do one full example together (like the coffee cup or waste example) before students attempt their own.

- **Keep Need Separate from Solution:**

Watch closely — students often write a solution disguised as a need (e.g., "needs a compost bin" instead of "needs a way to separate wet and dry waste").

- **Use the Canvas as a Cross-Check:**

After the canvas is filled, have students re-read their problem statement and ask, "Does every box support this statement?"

- **Push for Numbers:**

In Quantifiable Impact, insist on a real number or unit, not just a vague statement.

- **Encourage Referring Back:**

Remind students the canvas should draw directly from their earlier Activity Analysis and Empathy Map work, not new guesses.

Troubleshooting Points For Educators

1. Students write the "Need" as a solution instead of a need

Students jump ahead and describe a fix, instead of stating what the user actually needs.

Example: Writing "needs a smart bin with a sensor" (a solution) instead of "needs a simple way to separate wet and dry waste" (a need). Fix: Ask, "Are you telling me what they need, or how you'd solve it? Please don't mention any products yet."

2. Students confuse "Insight" with "Problem".

The Insight should explain why the need exists (the root cause); students often just restate the problem again. Example: Writing the Insight as "waste gets mixed" (restating the problem) instead of "because there's no awareness and no dustbin in every room" (the real cause). Fix: Ask, "You've told me what's happening — now tell me why it's happening in the first place."

3. Students leave the Problem Identification Canvas boxes too vague.

Short, generic answers like "sometimes" for Context or "everyone" for Customers don't give enough to work with. Example: Writing "people" under Customers instead of "families without a household help dedicated to sorting." Fix: Ask a follow-up for every box: "Which people, exactly? All the time, or in specific situations?"

4. Students mix up "Alternatives" with "Alternative Shortcomings".

Students describe what's wrong with a fix under Alternatives, instead of first stating what the current fix actually is. Example: Writing "doesn't work" under Alternatives instead of "buying an extra bin" (the alternative itself); "doesn't work" belongs under Shortcomings. Fix: Explain: "Alternatives = what do they do today? Shortcomings = why doesn't that fully solve it?"

5. Students skip the Quantifiable Impact box or leave it as a feeling

Since numbers are harder to find than opinions, students often write an emotional statement here instead of a measurable one. Example: Writing "it's really bad" under Quantifiable Impact instead of an estimate with a unit, like "amount of waste sent to landfill per week." Fix: Explicitly ask, "Can you attach a number or a unit to this — kg, rupees, hours, days missed?" Even a rough estimate is fine.

6. Students don't cross-check the canvas against their problem statement

After filling the canvas, students move straight to ideation without checking if the canvas actually supports their User-Need-Insight statement. Example: Problem statement mentions

"families," but the Customers box in the canvas says "shopkeepers" — a mismatch that weakens the whole definition. Fix: Before moving on, ask students to read their problem statement out loud, then check each canvas box against it one by one.

Tips For Using This Framework

- **Use the Sentence Formula:**

"<User> needs <need> because <insight>" keeps the problem statement tight and consistent across all groups.

- **Draw Directly from Earlier Tools:**

The canvas works best when filled using real material already gathered in Activity Analysis and the Empathy Map — not fresh guesses.

- **Keep Language Simple:**

Avoid jargon; the problem statement should be understandable to anyone reading it for the first time.

- **Revisit and Revise:**

It's normal to rewrite the problem statement 2-3 times as the canvas reveals new angles.

Benefits For Students

- **Builds Precision in Thinking:**

Trains students to separate needs, causes, and solutions instead of blending them together.

- **Strengthens Evidence-Based Reasoning:**

Requires real data (numbers, alternatives, emotional impact) instead of assumptions.

- **Prevents Wasted Effort:**

A well-defined problem stops students from designing solutions for the wrong issue.

- **Improves Written Communication:**

Practicing a clear, one-sentence problem statement is a valuable skill far beyond entrepreneurship.

06. SCAMPER

It is a structured ideation tool that uses 7 different "lenses" — Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse — to help students generate many possible solutions to a problem, instead of getting stuck on just one idea. SCAMPER pushes students to look at their problem from multiple creative angles before settling on a final solution.

Why Use The SCAMPER Framework?

- **Expands Idea Generation:**
Prevents students from stopping at their first (often obvious) idea.
- **Structured Creativity:**
Gives 7 clear starting points so "brainstorming" doesn't feel vague or overwhelming.
- **Builds on Existing Ideas:**
Instead of inventing from scratch, SCAMPER teaches students to twist, combine, or repurpose things that already exist.
- **Prepares for Prototyping:**
A wide pool of ideas from SCAMPER gives students real options to choose from before building an MVP.

How To Use The SCAMPER Framework?

1. Start with the Defined Problem:

Bring back the problem statement and insight from the Problem Identification Canvas.

Example: Families need a simple way to separate wet and dry waste at home, because of low awareness and missing dustbins in every room.

2. Apply Each SCAMPER Lens to the Problem:

- **Substitute:**
What can be swapped out for something else? Example: Substitute a single mixed dustbin with two color-coded bins (green for wet, blue for dry).
- **Combine:**
What can be merged with something else to create a new idea? Example: Combine a dustbin with a simple weighing scale to track how much waste is generated daily.
- **Adapt:**
What idea from another context/industry can be adapted here? Example: Adapt the "traffic light" system (red/yellow/green) used in factories to label bins by waste type.
- **Modify (Magnify/Minimize):**
What can be made bigger, smaller, or changed in some way? Example: Modify the bin size to be smaller and lighter so it fits in every room, even small ones.

- **Put to Another Use:**

Can this idea/object be used for a different purpose? Example: Put the segregated dry waste bin to another use — collected dry waste is exchanged for small rewards, encouraging correct sorting.

- **Eliminate:**

What can be removed to simplify the problem or solution? Example: Eliminate the need for a lid, since lids often stop people from bothering to segregate.

- **Reverse:**

What if the process was done in the opposite order or way? Example: Instead of the housekeeping staff segregating waste after collection, reverse it — segregation happens first, by the household, before the staff even arrives.

3. Document All Ideas:

Write down every idea generated under each lens — even ones that seem unusual or incomplete at this stage.

4. Shortlist the Strongest Ideas:

Review all SCAMPER ideas together and mark the 2-3 that best solve the core problem/ insight.

5. Move to Prototyping:

Take the shortlisted idea(s) forward into MVP/prototyping.

How Can Educators Use The SCAMPER Framework?

- **Explain Each Lens with a Simple Example:**

Use an everyday object (like a dustbin, water bottle, or school bag) to demonstrate all 7 lenses before students try their own problem.

- **Don't Let Students Skip Lenses:**

Some lenses (like Reverse or Eliminate) are harder — push students to attempt every one, even if the idea feels silly.

- **Encourage Quantity Over Quality First:**

Remind students the goal at this stage is many ideas, not perfect ones — judging comes later.

- **Guide Real-World Connections:**

Help students think of examples from things they've seen (apps, shops, machines) that used a similar SCAMPER-style twist.

- **Promote Group Brainstorming:**

Have groups shout out ideas quickly for each lens (a few minutes per lens) to keep energy and creativity high.

Troubleshooting Points For Educators

1. Students confuse "Substitute" with "Eliminate"

Students remove a part of the idea entirely and call it a substitution, instead of replacing it with something else. Example: Under Substitute, writing "no more dustbin" (this is Eliminate) instead of "replace one mixed dustbin with two color-coded bins" (this is Substitute). Fix: Clarify: "Substitute means swapping one thing for another. Eliminate means removing something completely, with nothing replacing it."

2. Students give the same idea repeated under different lenses

Students write nearly identical ideas for Modify, Adapt, and Combine, without really applying each lens differently. Example: Writing "smaller bin" under both Modify and Adapt, without adapting anything from another context for the Adapt lens. Fix: Ask, "What makes this idea specifically a Modify idea, and not an Adapt or Combine idea?" Push for a genuinely different angle per lens.

3. Students struggle with "Reverse" and leave it blank

Reverse is the hardest lens to grasp, so students often skip it or write something unrelated. Example: Leaving Reverse blank instead of trying "What if segregation happened before collection, not after?" Fix: Give a simple prompt: "What is the normal order of steps here? Now, what happens if we do the steps backward, or flip who does what?"

4. Students only generate "safe," obvious ideas

Students play it safe and avoid bold or unusual ideas, limiting the value of brainstorming. Example: Every lens produces some version of "make a poster to raise awareness," without exploring more creative options like combining, rewarding, or reversing steps. Fix: Push with a rule: "No idea is wrong at this stage — if it sounds strange, write it down anyway. We'll filter later."

5. Students jump to judging ideas too early

Instead of listing many ideas first, students stop after 1-2 ideas per lens because they're already deciding what's "good" or "bad." Example: A student writes one idea under Combine, then says "this is the best one, let's move on" without exploring 2-3 more options. Fix: Set a minimum target (e.g., at least 2 ideas per lens) before allowing any evaluation or shortlisting.

6. Students don't connect their SCAMPER ideas back to the original problem/insight

Some ideas sound creative but don't actually solve the core need identified earlier. Example: An idea about "colorful bin designs" is fun, but doesn't address the root insight — that people don't know why to segregate in the first place. Fix: After listing ideas, ask students to check each one against the problem statement: "Does this idea actually solve the need and insight we identified earlier?"

Tips For Using The SCAMPER Framework

- **Use a Simple Object First:**
Practice all 7 lenses on something familiar (a water bottle, school bag) before applying SCAMPER to the real problem.
- **Time-Box Each Lens:**
Give students 2-3 minutes per lens to keep the brainstorming fast and energetic, not overthought.
- **Keep a Running List:**
Write every idea on a chart or sticky notes so nothing is lost before the shortlisting step.
- **Revisit the Insight Often:**
Keep the original problem/insight statement visible while brainstorming, so ideas stay grounded in the real need.

Benefits For Students

- **Expands Creative Range:**
Teaches students that there isn't just one "right" solution — many angles can lead to good ideas.
- **Builds Structured Brainstorming Skills:**
Gives students a repeatable method they can use for any future problem, not just this one.
- **Encourages Risk-Taking in Ideas:**
Creates a safe space to suggest bold, unusual ideas without immediate judgment.
- **Prepares for Real Prototyping:**
Leaves students with a solid pool of tested, insight-backed ideas ready to be built and tested as an MVP.

07. Converting An Idea Into Features And Benefits

It is a structured tool to break down a business idea or product into its Features (what it actually has or does) and Benefits (what the user actually gains from it). This distinction helps students understand that customers don't buy features — they buy the benefits those features deliver.

Why Use This Framework?

- **Shifts Focus to the Customer:**
Trains students to think from "what my product has" to "what my customer gets."
- **Strengthens Pitching Skills:**
A benefit-led pitch is far more persuasive than a feature-led list.
- **Clarifies Value:**
Helps students see clearly why someone would actually choose their product.
- **Builds the Foundation for Value Proposition:**
This exercise feeds directly into writing a strong Unique Value Proposition (UVP) later.

How To Use This Framework?

1. Pick a Product or Idea:

Choose a real product, brand, or the student's own business idea. Example: A mattress/ cot product.

2. List the Features:

Features are the factual, technical details of the product — what it's made of, how it's built, its specifications. Example: Made of sag wood, 6.5 ft long, high tension springs, 2 inch coir.

3. Convert Each Feature into a Benefit:

For every feature, ask "So what does this actually give the customer?" and write the real-life benefit. Example:

- High tension springs + 2 inch coir → Sound sleep
- Made of sag wood → Durable
- 6.5 ft long, adjustable design → Body adjustable

4. Try a Second Example to Reinforce:

Example: Domino's Pizza

- **Feature:**
0-minute delivery guarantee (or free).
- **Benefit:**
Reliable, fast service the customer can trust and plan around — no waiting anxiety, pizza arrives fresh and hot.

5. Document Insights:

Lay out Features and Benefits side-by-side so students can see the direct link between each one.

6. Use in the Pitch:

When students later pitch their idea, coach them to always lead with the benefit, and use the feature only to support/prove it.

How Can Educators Use This Framework?

- **Use Familiar Brands First:**

Start with a well-known brand (like Domino's) before moving to the students' own idea — it's easier to spot the difference when the product is familiar.

- **Ask "So What?" Repeatedly:**

Whenever a student states a feature, keep asking "So what does that mean for the customer?" until they land on a real benefit.

- **Pair Every Feature with a Benefit:**

Don't let any feature sit alone — insist that each one is connected to a matching benefit.

- **Use a Two-Column Format:**

Have students physically write Features and Benefits in two separate columns to keep the distinction clear.

- **Promote Group Discussion:**

Let groups debate whether a listed point is really a feature or a benefit — it strengthens understanding through discussion.

Troubleshooting Points For Educators

1. Students list a feature but call it a benefit

Students often just restate a product detail and label it as a benefit, without translating it into a real-life gain. Example: Writing "2 inch coir" under Benefits, instead of translating it to "sound sleep" — the coir itself is a feature, not a benefit. Fix: Ask, "Is this something the product HAS, or something the customer GETS? Features are what it has, benefits are what they get."

2. Students give benefits with no matching feature

Students state impressive-sounding benefits ("makes life easier") but can't point to any actual feature that delivers it. Example: Writing "convenient" as a benefit without any feature (like fast delivery, one-click ordering) that actually creates that convenience. Fix: Ask, "What exact feature of the product makes this true? If you can't point to one, the benefit isn't backed up yet."

3. Students stop at the first "so what" instead of digging deeper

Students convert a feature into a shallow benefit and stop, missing the real emotional or practical payoff. Example: "High tension springs" → "comfortable" (shallow) instead of going one level deeper to "sound sleep" (the real, felt benefit). Fix: Push with a second "So what does THAT mean for them?" — often the best benefit is two steps away from the feature, not one.

4. Students confuse a Benefit with a Feature described positively

Sometimes students just add a positive adjective to a feature and think that makes it a benefit. Example: Writing "good high tension springs" as the benefit, instead of the actual outcome — "sound sleep." Fix: Explain: "A benefit is not a nicer-sounding feature. It's the actual result or feeling the customer experiences."

5. Students only think of functional benefits, missing emotional ones

Students focus purely on practical benefits (saves time, saves money) and miss emotional benefits (peace of mind, confidence, trust). Example: For Domino's "30-minute delivery," only writing "fast" (functional) and missing "no waiting anxiety, can trust and plan around it" (emotional). Fix: Ask, "Besides saving time or money, how does this make the customer FEEL?"

6. Students can't connect the Features/Benefits exercise to their own idea

Students do fine with the example (Domino's, mattress) but struggle to apply the same thinking to their own business idea. Example: A group can easily list Domino's benefits but leaves their own idea's Features/Benefits table empty or vague. Fix: Walk them through it step-by-step: first ask them to list 3-4 features of their own idea, then apply the "so what" question to each one individually.

Tips for Using This Framework

- **Use the "So What?" Test:**

This one question is the fastest way to check if something is a feature or a benefit.

- **Two-Column Layout:**

Keep Features on one side and Benefits directly opposite, so the connection is visually clear (as in the reference slides).

- **Start with Familiar Products:**

Everyday brands (Domino's, a mattress, a water bottle) make the concept click faster than abstract examples.

- **Always Lead with Benefits in a Pitch:**

Teach students that customers care about benefits first — features are the proof, not the headline.

Benefits For Students

- [illegible]

08. Minimum Viable Product (MVP)

It is a structured concept for building the simplest possible working version of an idea — just enough for early users to experience it, give feedback, and help refine it — before investing time and money into a fully finished product.



Why Use The MVP Framework?

- **Real Touch & Feel:**
Gives users something tangible to actually experience, instead of just hearing about the idea.
- **Easy to Get Feedback:**
A working version, even a basic one, makes it far easier to collect honest user feedback than a description or plan alone.
- **Frugal and Flexible:**
Stays low-cost and simple, so it's easy to change based on what feedback reveals.
- **Less Expensive to Build:**
Saves time, money, and effort by avoiding a fully polished product before knowing if the idea actually works.

How To Use The MVP Framework?

1. Start with the Shortlisted Idea:

Take the idea selected from SCAMPER/ideation that best solves the defined problem.

2. Break MVP into 3 Parts and Build Each One:

- **M — Minimum:**

Build the most rudimentary, bare-bones foundation of the solution possible — no extra features, no polish. *Example: For a waste segregation solution, this could just be two labelled cardboard boxes instead of a manufactured two-bin unit.*

- **V — Viable:**

Make sure it's sufficient enough for early adopters to actually use and understand what it does. *Example: The two boxes are clearly labelled "Wet" and "Dry" with simple pictures, so a household can genuinely use them for a week.*

- **P — Product:**

Ensure it's something tangible — customers can touch, feel, and interact with it, not just hear about it. *Example: A physical prototype placed in 2-3 real households, not just a drawing or description.*

3. Test with Real Users:

Give the MVP to a small group of real users (early adopters) and observe how they actually use it.

4. Collect Feedback:

Ask users what worked, what confused them, and what they'd change.

5. Modify Based on Feedback:

Since the MVP is frugal and simple, make quick, low-cost changes based on what was learned. *Example: If users found the boxes too flimsy, the next version could use light plastic crates instead — a small change, tested again.*

6. Repeat if Needed:

Keep testing and refining the MVP in small, inexpensive cycles before moving to a bigger investment or final product.

Troubleshooting Points For Educators

1. Students try to build a "final," polished product instead of an MVP

Students spend excessive time and materials perfecting the look of their prototype, missing the point of "minimum." *Example: Spending an entire session decorating and painting a waste-segregation bin instead of quickly testing if two labelled boxes work.* Fix: Ask, "Is this detail necessary to test your idea, or are we polishing something that isn't proven yet?" Redirect effort to function over form.

2. Students confuse MVP with the final idea/solution itself

Students think the MVP has to include every planned feature, rather than just enough to test the core idea. Example: Trying to add a weighing scale, reward system, and app all at once in the first version, instead of testing just the basic segregation idea first. Fix: Ask, "What is the ONE core thing we need to test right now? Everything else can wait for the next version."

3. Students skip real user testing

Groups build the MVP but only show it to each other, never testing it with an actual outside user. Example: A group is confident their bin design works, but has never actually asked a family member or classmate to try using it. Fix: Make testing with at least 2-3 real people (outside the group) a mandatory step before moving forward.

4. Students take feedback personally and resist changing their idea

Since students feel ownership over their idea, negative feedback can feel discouraging, leading them to ignore it. Example: A user says the box is too small, but the group insists their original size is fine without adjusting. Fix: Reframe feedback as data, not criticism: "This isn't about your idea being wrong — it's information that helps make it better."

5. Students confuse "Viable" with "impressive"

Students think viable means it has to impress people, rather than simply being usable enough for early feedback. Example: Adding decorative elements to make the MVP "look good" instead of focusing on whether it functions well enough to test. Fix: Clarify: "Viable just means someone can actually use it and give real feedback — it doesn't need to be impressive yet."

6. Students treat the MVP as a one-time build with no room for change

Students build one version and consider it "done," missing that MVP is meant to be modified based on feedback. Example: After testing, users mention the labels are confusing, but the group doesn't revise the design and moves straight to the final pitch. Fix: Build in a mandatory revision step: "Now that you have feedback, what is ONE change you will make before the next test?"

Tips For Using The MVP Framework

- **Keep Materials Cheap:**

Cardboard, paper, tape, and everyday items are ideal — spending little means it's easy to redo.

- **Test Early, Test Often:**

Don't wait for a "perfect" version to test — the earlier real users interact with it, the more useful the feedback.

- **Focus on the Core Function:**

An MVP should test the single most important part of the idea, not every feature at once.

- **Keep a Feedback Log:**

Have students note down exactly what users said, so changes are based on real input, not assumptions.

Benefits For Students

- **Builds Practical, Hands-On Thinking:**

Moves students from planning on paper to actually creating and testing something real.

- **Teaches Resourcefulness:**

Encourages students to build effectively with limited materials and budget.

- **Strengthens Resilience to Feedback:**

Helps students see feedback as a tool for improvement, not a judgment of their ability.

- **Prepares for Real Entrepreneurship:**

Mirrors how real businesses test ideas cheaply and quickly before large investments — a valuable, transferable skill.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

09. Building A Value Proposition

It is a structured way of defining why a customer should choose your product or service over every other option available — by identifying what makes it different, better, or more relevant to a specific customer's needs. A strong Value Proposition sits at the intersection of what customers need and what competitors are NOT already offering well.

Why Use This Framework?

- **Clarifies Differentiation:**

Forces students to identify exactly how their idea stands out from existing options.

- **Prevents "Me-Too" Products:**

Without a clear value proposition, a new idea just becomes another copy of what already exists.

- **Customer-Focused:**

Keeps the emphasis on why the customer benefits, not just what the product does.

- **Sets Up Pricing and Positioning:**

A clear value proposition later helps decide how to price and market the product.

How To Use This Framework?

1. **Study the Competitive Landscape First:**

Look at existing players in the same category and note what each one claims to offer.

Example: Toothpaste brands — Colgate claims "Maximum Cavity Protection," Pepsodent claims "Germi-Check, Fights Germs," Closeup claims "12-Hour Fresh Breath."

2. **Map Out Overlaps and Gaps:**

Compare what each competitor promises. Look for where their claims overlap (crowded space) and where there's a gap (unclaimed space). Example: In the pizza delivery space — Domino's is strongly associated with "30-minute delivery," Pizza Hut with "dine-in experience." Plotting competitors this way (as in the overlapping-lines diagram) shows where a new player could position itself differently — e.g., focusing on health-conscious pizza, which no big player owns yet.

3. **Identify What's Missing for the Customer:**

Ask what real customer need is still unmet by all current options. Example: Is there a value proposition gap in the grocery & vegetable market? Most sellers compete only on price or freshness — none may be solving for "guaranteed same-day delivery with zero substitution," which could be a real gap.

- 4 **Write the Value Proposition:**

Combine the customer's need with what makes your idea uniquely able to solve it, in one clear statement. Example: "For families who want fresh vegetables without last-minute substitutions, [idea] guarantees exact-order delivery every time — something no other local vendor currently promises."

5 Test the Value Proposition Against Competitors:

Check if the statement is something a competitor could easily copy or already offers — if so, it's not truly differentiated yet.

6 Refine Until It's Unique and Believable:

Keep sharpening the statement until it clearly answers "why choose us and not them?"

How Can Educators Use This Framework?

- **Use Familiar Categories First:**

Toothpaste or pizza brands work well since students already know multiple competitors in these categories.

- **Draw the Positioning Map Together**

Use a simple diagram (like the overlapping shapes) to visually show where competitors sit and where gaps exist.

- **Push Beyond the Obvious Claim:**

If a group's value proposition sounds like "we are the best," ask them to explain specifically what makes them the best, for whom.

- **Test for Copy-ability:**

Ask, "Could [Competitor X] just copy this claim tomorrow?" If yes, the value proposition isn't unique enough yet.

- **Encourage Real Research:**

Have students actually check labels, ads, or websites of 2-3 competitors before writing their own value proposition.

Troubleshooting Points For Educators

1. Students describe features instead of a value proposition

Students list what their product has, instead of explaining the specific advantage it gives over competitors. Example: Writing "our toothpaste has fluoride" instead of "our toothpaste is the only one specifically designed for sensitive teeth in hot climates." Fix: Ask, "Every competitor probably has this feature too — what makes YOUR version different or better for a specific customer?"

2. Students claim to be "better at everything"

Students write vague, all-encompassing claims that don't hold up to scrutiny. Example: "We are cheaper, faster, and better quality than everyone" — an unrealistic and unfocused claim. Fix: Ask, "If you had to pick just ONE thing you're truly best at, what would it be?" Push for focus over breadth.

3. Students don't research competitors before writing their value proposition

Without knowing what already exists, students risk repeating a claim a competitor already owns. Example: Claiming "fastest delivery" without realizing a competitor (like Domino's) has already built their entire brand around that exact promise. Fix: Require a quick competitor scan first — "List 2-3 things each competitor already claims, before writing your own value proposition."

4. Students confuse Value Proposition with a slogan or tagline

Students write something catchy but shallow, without real substance behind it. Example: "Freshness you'll love!" — catchy, but doesn't explain what specific need it solves or how it's different from competitors. Fix: Ask, "If a customer asked 'why should I believe that?' — what's your evidence or explanation?"

5. Students think value only means low price.

Students default to "we are the cheapest" as their entire value proposition, ignoring other forms of value. Example: Assuming a lower price alone is enough, without considering that value can come from convenience, quality, trust, or experience. Fix: Introduce the "Are they all priced the same?" question — showing that similar products (like different coffees) can have very different prices because of different types of value, not just cost.

6. Students' value proposition doesn't match their earlier Empathy Map/Problem insights

The written value proposition sounds good, but doesn't actually connect back to the real user need identified earlier. Example: Insight said "user needs to know coffee temperature," but the value proposition talks about "great taste" instead of addressing that specific need. Fix: Ask students to check: "Does this value proposition directly solve the insight/need we identified earlier? If not, revise it."

Tips For Using This Framework

Use a Competitor Map:

A simple visual (like overlapping shapes or a table) showing what each competitor claims makes gaps easier to spot.

Keep it to One Sentence:

A strong value proposition should be sayable in one clear, simple sentence.

Test with "Why Us and Not Them?":

This single question quickly reveals whether the value proposition is truly differentiated.

Revisit After Ideation:

As ideas evolve through SCAMPER and MVP testing, the value proposition may need to be rewritten to match the refined idea.

Benefits For Students

- **Builds Competitive Awareness:**

Trains students to look outward at the market, not just inward at their own idea.

- **Strengthens Strategic Thinking:**

Moves students from "what we built" to "why it matters more than the alternatives."

- **Improves Pitching:**

A clear value proposition becomes the core line used in every future pitch.

- **Prepares for Business Modelling:**

This exercise feeds directly into the Business Model Canvas and pricing decisions ahead.

10. How Do You Decide Value?

It is a structured way of understanding that "value" is not the same as "price" — the same basic product (like a cup of tea or coffee) can be priced very differently depending on the experience, brand, and context it's sold in. This concept helps students understand what customers are really paying for.

Why Use This Framework?

- **Separates Price from Value:**
Helps students see that price is just a number, but value is what the customer feels they're getting for that number.
- **Explains Pricing Differences:**
Answers the common student question — "why do people pay more for basically the same thing?"
- **Guides Smart Pricing Decisions:**
Helps students set a price for their own idea based on the value it delivers, not just the cost of materials.
- **Builds Market Awareness:**
Trains students to notice how brands add non-physical value (experience, trust, status) beyond the product itself.

How To Use This Framework?

1. **Compare the Same Basic Product Across Different Sellers:**
Pick one simple product sold in very different settings. Example: Chai from a roadside stall, coffee from a café, and coffee from Starbucks.
2. **Ask "Are They All Priced the Same?":**
Get students to guess or research actual prices for each version. Example: Roadside chai might cost ₹10-15, café coffee ₹80-100, Starbucks coffee ₹250-300 — despite all being variations of a hot beverage.
3. **Identify What's Different Beyond the Product Itself:**
List the non-product factors that justify the price difference. Example: Starbucks offers air-conditioned seating, consistent taste worldwide, brand status, comfortable ambience, and free wifi — none of which the roadside stall offers.
4. **Connect Value to the Customer's Perception, Not Just Cost:**
Explain that value is decided by what the customer believes they're getting, not just what it costs to make. Example: The actual coffee/milk cost is a small fraction of the Starbucks price — the rest is priced based on experience and brand value.
5. **Apply This to the Student's Own Idea:**
Ask what non-physical value (convenience, trust, experience, exclusivity) their own idea could offer, beyond just the raw product/service.

6. Set Pricing Based on Total Value, Not Just Cost:

Guide students to price their idea based on the full value delivered, not only the material cost.

How Can Educators Use This Framework?

- **Use Relatable Daily Examples:**

Chai vs. café coffee vs. Starbucks works well because most students have direct experience with at least one of these.

- **Ask "What Would You Pay For?":**

Get students to personally reflect on what would make them pay more for a similar product.

- **Highlight Invisible Value:**

Point out factors like ambience, trust, and consistency that aren't part of the product itself but still affect price.

- **Avoid Judging One as "Better":**

Make clear that no option is inherently right or wrong — each is priced for a different kind of value and different customer.

- **Link Back to Value Proposition:**

Show students how the value proposition they wrote earlier directly explains what price their idea could reasonably charge.

Troubleshooting Points For Educators

1. Students think value is only about ingredients/materials.

Students assume higher price only makes sense if the raw product is genuinely better quality. Example: Assuming Starbucks coffee must use significantly better coffee beans to justify its price, without considering brand, ambience, and consistency. Fix: Ask, "Besides the coffee itself, what else are you paying for when you sit at Starbucks?"

2. Students believe cheaper always means better value.

Students equate "value" directly with "lowest price," missing that value depends on what the customer wants. Example: Saying roadside chai is "better value" for everyone, without considering that some customers specifically want the ambience or consistency Starbucks offers. Fix: Ask, "Better value for WHOM? A student in a hurry, or someone having a meeting with a client?" Different customers value different things.

3. Students can't explain why they'd personally pay more for something.

When asked to price their own idea, students default to the lowest possible number without connecting it to value delivered. Example: Pricing their innovation at the bare material cost, ignoring the convenience or problem it solves for the customer. Fix: Ask, "If your idea saves someone an hour, or solves a real frustration — is that worth something beyond just the materials?"

4. Students confuse "expensive" with "valuable".

Students assume a higher price alone proves something is more valuable, without evidence. Example: Assuming a costly product must be better, without checking if the price is actually justified by real added value. Fix: Ask, "Is this expensive because it delivers more value, or just because of the brand name? How would you check?"

5. Students struggle to apply this concept to their own idea.

Students understand the coffee example but can't translate the concept of "hidden value" into pricing their own product. Example: A group can explain why Starbucks costs more, but sets their own idea's price using only material cost, ignoring convenience or experience they offer. Fix: Walk through it directly: "What does your idea offer besides the basic product — time saved, less worry, better experience? Does that deserve a higher price?"

6. Students think all customers value the same things equally.

Students assume every customer would choose the same option, missing that value is subjective and depends on the customer's context. Example: Assuming everyone would always choose the cheapest chai over Starbucks, without considering occasions (meetings, work calls) where the Starbucks experience matters more. Fix: Ask, "Can you think of a situation where YOU might choose the more expensive option? Why?"

Tips For Using This Framework

- **Use Visual Price Comparisons:**

Showing actual images and prices side-by-side (like chai vs. café vs. Starbucks) makes the concept click faster than just explaining it.

- **Connect to Real Personal Experience:**

Ask students to recall a time they personally paid more for something similar — it makes the concept concrete.

- **Separate "Cost to Make" from "Price to Sell":**

Keep these two ideas clearly distinct throughout the discussion.

- **Revisit During Business Modelling:**

This concept becomes directly useful when students later decide their own pricing in the Business Model Canvas.

Benefits For Students

- **Builds Real-World Business Sense:**

Helps students understand pricing decisions they see everywhere in daily life.

- **Strengthens Customer Empathy:**

Reinforces that different customers have different needs and are willing to pay differently for them.

- **Prepares for Pricing Decisions:**

Gives students a practical way to think about pricing their own idea later, based on value rather than guesswork.

- **Encourages Critical Observation:**

Trains students to notice pricing and value differences in everyday products around them.

11. Business Model Canvas

PROBLEM <i>List your top 1-3 problems.</i>	SOLUTION <i>Outline a possible solution for each problem.</i>	UNIQUE VALUE PROPOSITION <i>Single, clear, compelling message that states why you are different and worth paying attention.</i>	UNFAIR ADVANTAGE <i>Something that cannot easily be bought or copied.</i>	CUSTOMER SEGMENTS <i>List your target customers and sizes.</i>
EXISTING ALTERNATIVES <i>List how these problems are solved today.</i>	KEY METRICS <i>List the key numbers that tell you how your business is doing.</i>		CHANNELS <i>List your path to customers (direct or indirect).</i>	
COST STRUCTURE <i>List your fixed and variable costs.</i>		REVENUE STREAMS <i>List your sources of revenue.</i>		

It is a structured, one-page tool that maps out all 9 essential building blocks of a business — from what it offers, to who it serves, to how it makes money — so students can see their entire business idea at a glance before executing it.

Why Use The Business Model Canvas Framework?

- **Complete Business Overview:**
Captures the whole business on a single page instead of scattered notes.
- **Reveals Missing Pieces:**
Makes it obvious if a student has planned the product but forgotten how it will actually earn revenue, or vice versa.
- **Connects All Elements:**
Shows how each block (like Costs or Channels) directly affects others (like Revenue).
- **Practical and Visual:**
Uses simple boxes and prompts, making a complex business easier to plan and communicate.

How To Use The Business Model Canvas Framework?

1. Start with the Value Proposition (Center):

State clearly what problem you solve or what value you offer, and to whom. Example: MittiCool — an eco-friendly clay refrigerator that keeps food and water cool without electricity, for households with limited or unreliable power access.

2. Fill In All 9 Blocks:

- **Value Proposition:**

What value do you deliver to the customer? Example: Affordable, electricity-free cooling using natural clay technology.

- **Customer Segments:**

Who are your customers? Example: Rural and semi-urban households, especially in areas with frequent power cuts.

- **Customer Relationships:**

How do you interact with and retain your customers? Example: Direct trust-building through demonstrations at local fairs and word-of-mouth from satisfied users.

- **Channels:**

How does your product/service reach the customer? Example: Local retail stores, exhibitions, and direct village-level distribution.

- **Key Activities:**

What must you actually do to deliver the value proposition? Example: Sourcing quality clay, molding and firing the product, quality-testing each unit.

- **Key Resources:**

What resources are essential to make this work? Example: Skilled clay artisans, a kiln/firing facility, and raw clay supply.

- **Key Partners:**

Who do you rely on outside your own business? Example: Local clay suppliers, retail distributors, and NGOs supporting rural outreach.

- **Revenue Streams:**

How does the business earn money? Example: Direct sales of the clay refrigerator units.

- **Cost Structure:**

What are the major costs involved in running the business? Example: Raw material (clay), artisan wages, kiln fuel/electricity, transportation to retail points.

3. Check for Consistency:

Make sure every block supports the Value Proposition - for example, the Channels chosen should actually be able to reach the Customer Segments identified.

4. Document Insights:

Keep the filled canvas visible as a single-page reference for the whole business idea.

5. Use it to Guide the Pitch:

The filled canvas becomes the backbone of the final business pitch — each block gives one clear talking point.

How Can Educators Use The Business Model Canvas Framework? _____

- **Always Start from the Center:**

Insist students define the Value Proposition first — every other block should connect back to it.

- **Use a Real, Familiar Example:**

Walking through MittiCool (or a similarly relatable Indian brand) before their own idea makes the abstract blocks concrete.

- **Check Block-to-Block Logic:**

Ask connecting questions like "Does this Channel actually reach this Customer Segment?" to catch mismatches.

- **Don't Let Students Skip Cost/Revenue:**

These two blocks are often rushed — push students to be specific and realistic here.

- **Use Sticky Notes for Group Work:**

Let groups write ideas on sticky notes per block so they can easily move, discuss, and revise them together.

Troubleshooting Points For Educators _____

1. Students confuse Value Proposition with Key Activities

Students describe what they DO, instead of what value the customer actually GETS. Example: Writing "we make clay refrigerators" (this is a Key Activity) as the Value Proposition, instead of "electricity-free cooling for households without reliable power" (the actual value delivered). Fix: Ask, "Is this something you do, or something the customer gains? Value Proposition is always about the customer's benefit."

2. Students write vague Customer Segments

Students write "everyone" or "all people" instead of a specific, targeted group. Example: Writing "people who need refrigerators" instead of "rural households in areas with frequent or no electricity access." Fix: Ask, "Who specifically benefits most? Think about location, income, or lifestyle — not just everyone."

3. Students mix up Channels with Customer Relationships

Channels are about HOW the product reaches customers; Customer Relationships are about HOW you interact and build trust with them. Example: Writing "local demonstrations" under Channels, when it's really about building trust/relationship, while "retail stores" (distribution) belongs under Channels. Fix: Clarify: "Channels = the path the product travels to reach them. Relationships = how you connect with and keep them as customers."

4. Students leave Key Partners and Key Resources too generic

Students write broad terms like "suppliers" or "materials" without specifying what's actually needed. Example: Writing "materials" under Key Resources instead of "raw clay, kiln facility, skilled artisans." Fix: Ask, "What EXACT resource or partner does this depend on? Name it specifically."

5. Students underestimate or skip Cost Structure

Students only think about the selling price and forget to map out what it actually costs to run the business. Example: Filling in Revenue Streams in detail, but leaving Cost Structure blank or just writing "some cost." Fix: Ask, "What do you have to spend money on to make this happen — materials, labor, transport, rent? List each one."

6. Students don't check if the 9 blocks connect logically.

Students fill in each block separately without checking whether they support each other, creating contradictions. Example: Customer Segment says "low-income rural households," but Channels lists "high-end city malls" — a clear mismatch. Fix: After filling in all blocks, ask students to trace the connections aloud: "Does your Channel actually reach your Customer Segment? Does your Revenue Stream match what your Customer would realistically pay?"

Tips For Using The Business Model Canvas Framework

- **Always Anchor to the Value Proposition:**

Every other block should be checked against it — "Does this support what we promised to deliver?"

- **Use Visuals or Sticky Notes:**

A large printed canvas with movable sticky notes makes group collaboration and revision much easier.

- **Keep Each Point Short:**

One clear phrase per point is better than long paragraphs — the canvas should be scannable at a glance.

- **Revisit After Testing the MVP:**

Real feedback from MVP testing often reveals the Business Model Canvas needs small but important updates.

Benefits For Students

- **Builds Big-Picture Thinking:**

Helps students see their entire business as one connected system, not isolated ideas.

- **Reveals Real Gaps Early:**

Surfaces missing pieces (like unclear revenue or unrealistic costs) before real money or effort is spent.

- **Strengthens Pitching Skills:**

Each of the 9 blocks becomes a clear, structured talking point for the final pitch.

- **Prepares for Real Entrepreneurship:**

The Business Model Canvas is a globally used tool — learning it gives students a practical, transferable business skill.

12. Prototyping

Types Of Prototyping

Low-Fidelity Prototypes:

- **Definition:**
Simple, rough models used to quickly test and communicate ideas.
- **Characteristics:**
Inexpensive, fast to create, focuses on functionality rather than details.
- **Examples:**
Paper sketches, cardboard models, mock-ups made with simple materials like clay or tape.
- **Purpose:**
To validate concepts and gather initial feedback.

High-Fidelity Prototypes:

- **Definition:**
Detailed, closer-to-final models that look and function like the finished product.
- **Characteristics:**
More time and resources required, polished appearance, incorporates detailed features.
- **Examples:**
3D-printed objects, interactive digital apps, 3D mechanical designs.
- **Purpose:**
To test the functionality and usability of the nearly final design.

Prototyping Tips For Students

1. Start Small:

Focus on one aspect of the solution and expand later. Use low-fidelity prototypes first, once they start working, start with high-fidelity prototypes.

2. Use Available Materials:

Look for materials at home or in the classroom. Don't spend too much time and money on prototyping in the beginning.

3. Iterate Frequently:

Test and improve based on feedback. Fail fast by gaining feedback and keep improving your model. Test with actual users if possible.

4. Involve Peers:

Share your prototype with friends or classmates to get different perspectives.

5. Document the Process:

Take photos or videos to show how the prototype was made and tested.

Methods And Tools For Prototyping

Method	Description	Examples	Tools
Basic Prototyping	Using basic materials to create physical mock-ups.	• Paper sketches • Clay models • Cardboard structures	Tools: Scissors, glue, tape, colored paper, markers, cardboards, papers, clay, etc.
No-Code Prototyping	Creating models with no-code digital tools or using no technical tools.	• Wireframes • Flipbook animations • Storyboards • App/ website mock-ups	Non-tech Tools: Rulers, sticky notes, whiteboards, pencils, index cards. Tech tools: Software like Figma, Google Sites, Storyboard That, Canva, basic animation tools, etc.
Tech Prototyping	Incorporating technology to build interactive models.	• Virtual simulations • Simple robots	Tools: Tinkercad, Scratch, Arduino, Micro:bit, 3D printers, Makey Makey, etc.
Mixed Prototyping	Combining physical and digital elements. (Electronics/ Embedded Systems)	• Physical model with electronic components (e.g., motors)	Tools: Arduino boards, Raspberry Pi, 3D modeling software (e.g., Blender), soldering kits, IoT kits.
		• Smart devices with sensors	

Examples Of Prototypes: _____

Prototype	Description	Materials/Tools	Purpose
Low-Cost Water Filter (Non-Tech)	A physical model demonstrating water filtration using simple materials.	Plastic bottle, sand, gravel, cloth, small stones	To show how layered filtration can remove dirt from water.
Educational Game for Water Awareness (Basic materials)	A board game that teaches water conservation.	Cardboard, markers, dice, tokens	To engage students and raise awareness about water-saving techniques.
App for Clean Water Tracking (Digital/ Tech)	A simple app prototype for tracking clean water sources.	Figma (for interface design), Scratch (for interactive simulation)	To demonstrate how technology can help families find safe drinking water nearby.
Smart Irrigation System (Mixed)	A small prototype using sensors and Arduino to automate water use in farming.	Arduino board, moisture sensors, small water pump, plant pots	To showcase a cost-effective way to save water in agriculture.
Awareness Campaign Kit (Non-Tech)	A physical kit with posters, flyers, and activity sheets about waterborne diseases.	Paper, markers, string for banners	To educate communities through visual materials.

13. Elevator Pitch

It is a structured way of condensing an entire business idea into a short, powerful pitch — covering 6 key components (Value Proposition, Customer, Problem, Purpose, Economics, Vision) — that can be delivered clearly and confidently in the time it takes to ride an elevator with someone important.

Why Use The Elevator Pitch Framework?

- **Forces Clarity:**
Compresses a complex business idea into its most essential, understandable points.
- **Tests Real Understanding:**
If students can't explain their idea in a short pitch, it usually means something in their plan isn't clear yet.
- **Builds Confidence:**
Prepares students to speak about their idea to anyone — investors, judges, customers — at any moment.
- **Covers the Whole Business Quickly:**
Touches on problem, customer, value, purpose, money, and future scale—all in one short pitch.

How To Use The Elevator Pitch Framework?

1. Gather Material from Earlier Tools:

Pull directly from the Problem Identification Canvas, Value Proposition, and Business Model Canvas already completed.

2. Answer the 6 Guiding Questions, One per Component:

- **Value Proposition - What is so exciting about your offering?**
State the unique, compelling reason someone should care about this idea. Example: "Our device gives every household clean drinking water without needing electricity."
- **Customer — Who is going to buy/consume?**
Name the specific customer this is built for. Example: "Rural and semi-urban households with unreliable access to clean water."
- **Problem — What problem is it solving? What value is it adding to the consumer?**
State the real problem and the value delivered because of it. Example: "Families currently rely on boiling water or risky sources, leading to health issues and wasted time. Our solution gives them safe water instantly, saving both health and time."
- **Purpose — Why do you think you are best equipped to make this offering?**
Why are you doing this? Explain the personal or team reason behind the idea, and why you're the right person/team to solve it. Example: "Our team studied local water quality issues firsthand and worked with local artisans to design something affordable and easy to use."

- **Economics — How will you make money?**

Clearly explain the revenue model. Example: "We sell each filtration unit directly to households at an affordable price, with a small recurring cost for filter replacement."

- **Vision — How big do you think this venture can be?**

Paint the future scale — geography, number of customers, countries, revenue, profits. Example: "We aim to reach 10,000 households across our state in 2 years, and eventually expand to other regions facing similar water challenges."

3. Combine All 6 Answers into One Flowing Pitch:

String the answers together into 5-6 natural sentences, not a list.

4. Time It:

Practice saying the full pitch in under 60 seconds — the length of a real elevator ride.

5. Refine for Clarity and Energy:

Cut any jargon or long explanations; keep language simple, confident, and easy to follow.

6. Practice Out Loud:

Rehearse the pitch multiple times until it sounds natural, not memorized or robotic.

How Can Educators Use The Elevator Pitch Framework?

- **Use the 6 Questions as a Checklist:**

Have students answer each question separately in writing before combining them into a pitch.

- **Model One Full Example First:**

Deliver a sample elevator pitch (using a familiar idea) so students hear what a complete, well-paced pitch sounds like.

- **Time Every Practice Round:**

Use a stopwatch or timer app so students get used to condensing their pitch within 45-60 seconds.

- **Push for Natural Speech, Not Reading:**

Encourage students to practice enough that they aren't just reading from paper.

- **Give Specific Feedback per Component:**

Instead of vague feedback like "make it better," point out exactly which of the 6 components (e.g., Economics) needs more clarity.

Troubleshooting Points For Educators

1. Students confuse "Value Proposition" with "Problem"

Students describe the problem again under Value Proposition, instead of stating what makes their solution exciting. Example: Writing "clean water is a big problem" under Value Proposition (this is Problem) instead of "our device gives clean water without electricity" (this is the exciting offering). Fix: Ask, "Is this describing what's WRONG (Problem), or what's EXCITING about your fix (Value Proposition)?" Keep them as two separate, distinct sentences.

2. Students give a vague answer for "Customer"

Students say "everyone" or "the public" instead of naming a specific group. Example: Writing "people who need water" instead of "rural households with unreliable electricity and water access." Fix: Ask, "Who specifically would buy this first? Think about location, income level, or lifestyle."

3. Students skip or rush through "Purpose" Since Purpose

feels personal, students often leave it as a single generic line like "we care about people."

Example: Writing "we want to help society" instead of "our team researched local water issues directly and built a solution with local artisans." Fix: Ask, "What specifically makes YOUR team able to solve this — research, experience, connections, or a personal story?"

4. Students can't clearly explain "Economics."

Students struggle to state simply how the business earns money, often rambling or listing too many revenue ideas at once. Example: Listing 4-5 different possible revenue models instead of clearly stating the primary one. Fix: Ask, "If you had to name just ONE main way this makes money, what would it be? Start there."

5. Students make "Vision" unrealistic or too vague.

Students either give an overly grand claim ("we'll be in every country") or too small a vision, without real numbers or scope. Example: Saying "we'll be huge someday" instead of "reach 10,000 households in our state within 2 years." Fix: Prompt with the specific vision categories: "How many customers? Which geography? What timeframe? Give at least one real number."

6. Students read the pitch instead of speaking it naturally, or go far over time.

Students write a great pitch on paper but stumble or read monotonously when saying it aloud, or the pitch runs far too long. Example: A written pitch that takes 3 minutes to read aloud, when it needs to fit in under a minute. Fix: Have students practice multiple times with a timer, cutting one sentence at a time until it fits naturally within 45-60 seconds.

Tips For Using The Elevator Pitch Framework

- **Keep One Sentence per Component:**

A tight pitch usually has just one strong sentence for each of the 6 parts.

- **Practice Out Loud, Not Just on Paper:**

Reading it silently isn't the same as saying it confidently - practice speaking it multiple times.

- **Use a Timer:**

Keep every practice round under 60 seconds to build the discipline of concise speaking.

- **End with Vision for Impact:**

Save the "big picture" Vision line for last - it leaves the listener with an exciting, memorable final thought.

Benefits For Students

- **Builds Clear Communication:** Trains students to explain complex ideas simply and confidently, a skill useful far beyond entrepreneurship.
- **Strengthens Idea Ownership:** Forces students to fully understand their own idea well enough to summarize it quickly.
- **Prepares for Real Pitching Situations:** Mirrors exactly what's expected in investor meetings, competitions, or public speaking moments.
- **Boosts Confidence:** Practicing a tight, well-structured pitch multiple times builds real confidence in public speaking.



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